

Graph Theory & Combinatorics — Short Revision Notes (MATH)

Mathematics (MATH) · Unit 5 · Graph Theory & Combinatorics · 2025

Overview

These notes cover **Graph Theory & Combinatorics**. Engineering and discrete mathematics — linear algebra, calculus, probability, logic and graph theory for computer science. Each section states the core idea, a worked intuition and the points examiners look for.

1. Permutations and Combinations

nPr counts ordered arrangements and nCr counts unordered selections.

Why it matters: Permutations and Combinations appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of permutations and combinations in one line.
- A small example showing permutations and combinations in practice.
- Common mistakes students make and how to avoid them.

2. Pigeonhole Principle

If $n + 1$ objects go into n boxes, some box holds at least two objects.

Why it matters: Pigeonhole Principle appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of pigeonhole principle in one line.
- A small example showing pigeonhole principle in practice.
- Common mistakes students make and how to avoid them.

3. Recurrence Relations

Linear recurrences are solved with characteristic equations or generating functions.

Why it matters: Recurrence Relations appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of recurrence relations in one line.
- A small example showing recurrence relations in practice.
- Common mistakes students make and how to avoid them.

4. Euler and Hamiltonian Paths

Euler paths traverse every edge once; Hamiltonian paths visit every vertex once.

Why it matters: Euler and Hamiltonian Paths appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of euler and hamiltonian paths in one line.
- A small example showing euler and hamiltonian paths in practice.
- Common mistakes students make and how to avoid them.

5. Graph Colouring

The chromatic number is the minimum colours needed so adjacent vertices differ.

Why it matters: Graph Colouring appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of graph colouring in one line.
- A small example showing graph colouring in practice.
- Common mistakes students make and how to avoid them.

6. Planar Graphs

Euler's formula $V - E + F = 2$ holds for connected planar graphs.

Why it matters: Planar Graphs appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of planar graphs in one line.
- A small example showing planar graphs in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Permutations and Combinations · Pigeonhole Principle · Recurrence Relations · Euler and Hamiltonian Paths · Graph Colouring · Planar Graphs